

Normalization: A method for designing a relational database, is to use a process commonly known as normalization.

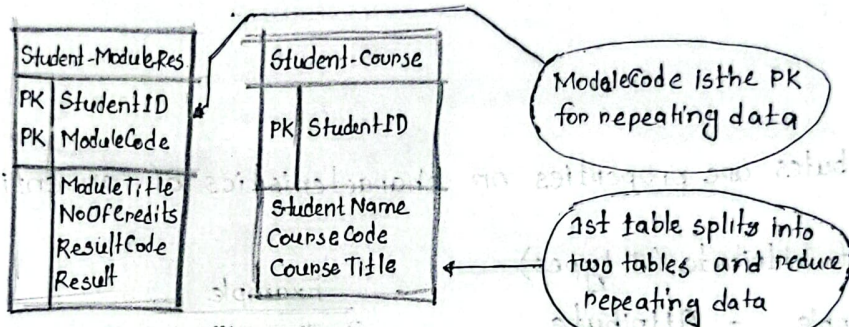
Why we use

- Reduce data redundancy
- No loss of information
- Save database space.

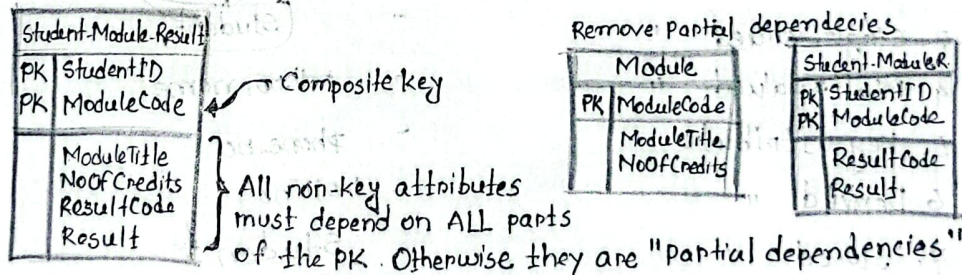
6 Types of normalization but basically use 3 forms

- 1st NF - remove repeating group of data
- 2nd NF - " partial dependency
- 3rd NF - " transitive dependency

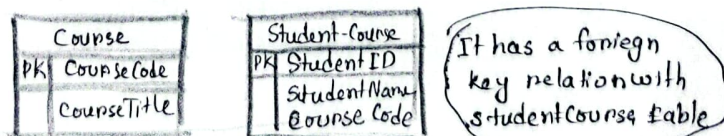
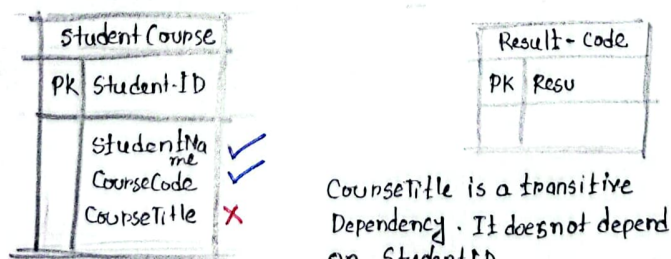
1st NF



2nd NF



3rd NF



KEY

- A key in DBMS is an attribute or a set of attributes that help to uniquely identify a tuple in a relation (table).

7 types of key

1. Primary key
2. Super key
3. Candidate key
4. Alternate key
5. Foreign key
6. Composite key
7. Unique key

Example

StudentID
OrderID, (OrderID, Orderdate)
Email, Phone
Phone
CustomerID (from Order table)
(StudentID, CourseID)
Email

Attribute

- Attributes are properties or characteristics of an entity.

Types of attribute (8 types)

1. Simple - Attribute
2. Composite "
3. Single-valued "
4. Multi-valued "
5. Key attribute
6. Derived "
7. Stored "
8. Complex "

Example

roll-no, class, age

Student

roll-no, name

phone-no

roll-no

Student

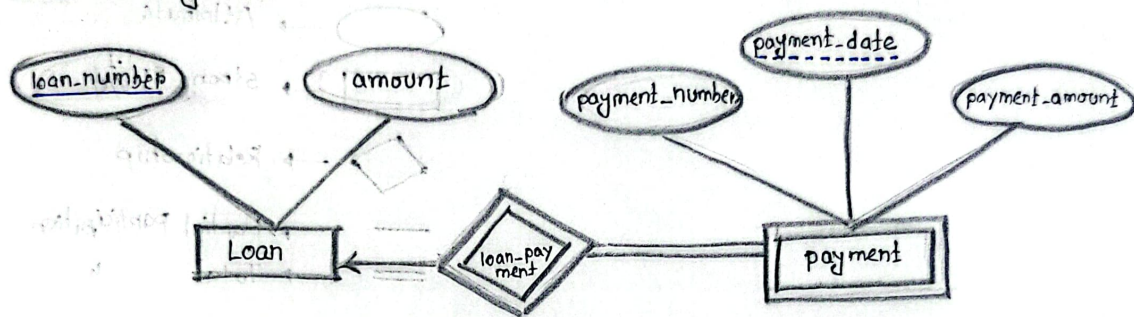
name

Entity sets

In a DBMS, an entity set refers to a collection of similar entities, essentially a group of related objects that belong to the same entity type.

Strong & weak entity sets

Weak entity set : An entity set may not have a sufficient attributes to form a primary key. Such an entity set is termed as weak entity set.

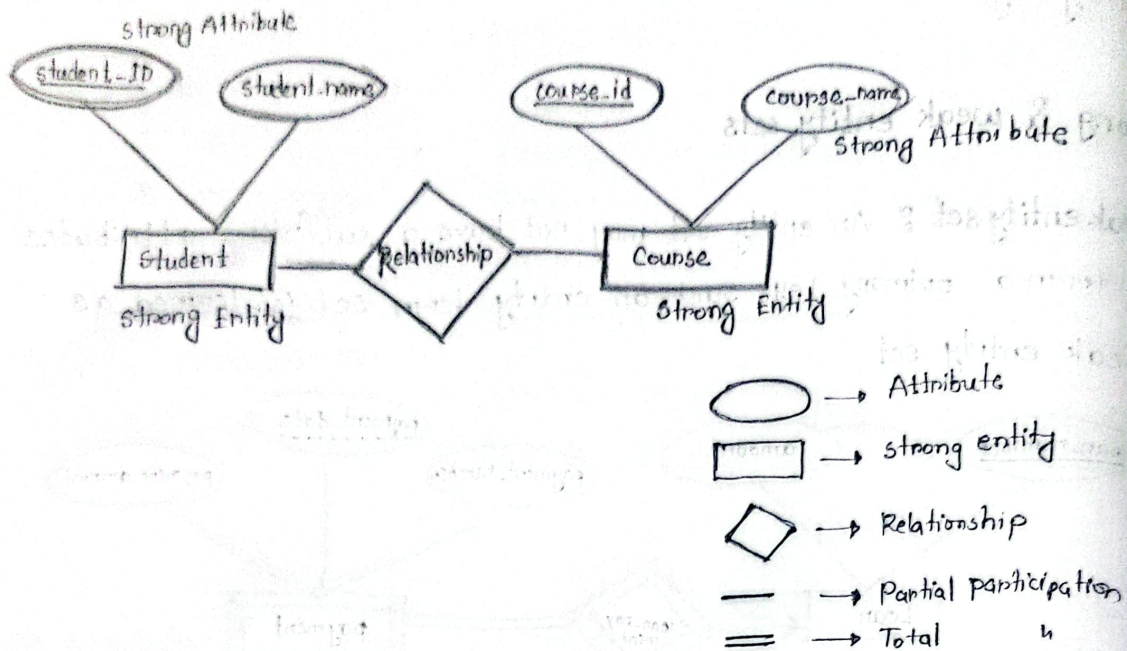


The existence of a weak entity set depends on the existence of a identifying entity set.

- (i) It must relate to the identifying entity set via a total, one to many relationship set from the identifying to the weak entity set.
- (ii) Identifying entity relationship depicted using a double diamond.
- (iii) **Discriminator (partial key) :** of a weak entity set of attributes that distinguishes among all the entities. e.g: payment-number.
- (iv) **Primary key of a weak entity set :**
pk of the strong entity set + weak entity set's discriminator

Strong Entity

is independent of any other in the schema. A strong entity is nothing but an entity set having a primary key.



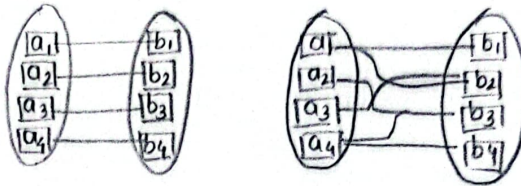
Strong Entity	Weak entity
1. Strong entity always has a primary key.	1. It will not have a PK but It has partial discriminator key
2. It is not dependent on any other entity.	2. is dependent on the strong entity.
3. Represented by a single rectangle	3. Represented by double rectangle
4. Relationship between two strong entities is represented by single diamond.	4. Relationship between a strong entity and a weak entity is represented by double diamond.
5. A strong entity may or may not have total participation	5. It has always total participation

Mapping Cardinalities

- Mapping cardinalities, or cardinality ratios, express the number of entities to which another entity can be associated via relationship set.

Types

- (i) One-to-one
- (ii) One-to-many
- (iii) Many-to-one
- (iv) Many-to-many



Relationship Set

- A relationship set is a set of relationships of the same type.